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Extendable platform lift assembly

Abstract

An extendable lift assembly for a vehicle, including a sliding rail assembly, including a plate slidably connected to the vehicle, and a knuckle arm pivotably connected to the plate, a first platform pivotably connected to the knuckle arm, and a second platform extending from and pivotably connected to the plate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is filed under 35 U.S.C. § 120 as a continuation of U.S. patent application Ser. No. 17/020,986, filed on Sep. 15, 2020, which application claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Patent Application No. 62/902,642, filed Sep. 19, 2019, which applications are incorporated herein by reference in their entireties.

FIELD

(1) The present disclosure relates to the field of lifts, and more particularly, to an extendable lift for entering and exiting a vehicle.

BACKGROUND

(2) Platform lifts for vehicles are well known for accommodating wheelchairs. Specifically, platform lifts for vehicles are designed carry a user seated in a wheelchair onto and off of a vehicle, such as a van. However, current lift designs are bulky and take up a large amount of space within the vehicle. Additionally, current lift designs don't account for standing users, or users who are not seated in a wheelchair but need assistance entering and exiting a vehicle so as to avoid dangerous steps.

(3) Thus, there is a long felt need for a platform lift assembly that extends out of a vehicle opening, such as a doorway, to retrieve a user and carries the user into the vehicle and can be stowed away

therein.

SUMMARY

(4) According to aspects illustrated herein, there is provided an extendable lift assembly for a vehicle, comprising a sliding rail assembly, including a plate slidably connected to the vehicle, and a knuckle arm pivotably connected to the plate, a first platform pivotably connected to the knuckle arm, and a second platform extending from and pivotably connected to the plate.

(5) In some embodiments, the extendable lift assembly further comprises a static rail operatively arranged to be fixedly secured to the vehicle, wherein the plate is slidably connected to the static rail. In some embodiments, the extendable lift assembly further comprises a motor including a gear, wherein the motor and the gear are operatively arranged to displace the plate with respect to the static rail. In some embodiments, the extendable lift assembly further comprises a rack connected to the plate, wherein the gear is operatively arranged to engage the rack. In some embodiments, the extendable lift assembly further comprises a lever operatively arranged to engage and disengage the gear from the rack.

(6) In some embodiments, the first platform is connected to the knuckle arm via a shaft. In some embodiments, the first platform is further connected to the knuckle arm via a rod, the rod being pivotably connected to the knuckle arm and pivotably and slidably connected to the first platform. In some embodiments, the knuckle arm is connected to the plate via a shaft. In some embodiments, the first platform is rotatable in a first circumferential direction relative to the knuckle arm and a second circumferential direction, opposite the first circumferential direction, and the knuckle arm is rotatable in a third circumferential direction relative to the plate and a fourth circumferential direction, opposite the third circumferential direction.

(7) In some embodiments, the extendable lift assembly further comprises a bridge plate hingedly connected to the first platform. In some embodiments, the extendable lift assembly further comprises an actuator connected at a first end to the bridge plate and at a second end to the first platform, the actuator operatively arranged to circumferentially displace both the bridge plate and the first platform. In some embodiments, the extendable lift assembly further comprises a locking plate slidably connected to the plate, and a pin connected to the knuckle arm, wherein the locking plate is operatively arranged to engage the pin to non-rotatably connect the knuckle arm and the plate.

(8) In some embodiments, the knuckle arm comprises an arm pivotably connected thereto, and the static rail comprises a slot, wherein the arm is operatively arranged to engage the slot to circumferentially displace the knuckle arm. In some embodiments the second platform is a seat for a user. In some embodiments, the extendable lift assembly further comprises a drive mechanism operatively arranged to displace the plate with respect to extend and retract the plate.

(9) According to aspects illustrated herein, there is provided an extendable lift assembly for a vehicle operatively arranged to extend from proximate a floor of the vehicle to proximate a ground surface, the extendable lift assembly comprising a static rail operatively arranged to be fixedly secured to the vehicle, a sliding rail assembly, including a plate slidably connected to the static rail, and a knuckle arm pivotably connected to the plate, a first platform pivotably connected to the knuckle arm, a second platform pivotably connected to the plate, and a drive mechanism operatively arranged to displace the sliding rail assembly with respect to the static rail.

(10) In some embodiments, at least one of the first platform and the second platform is operatively arranged to carry a user into and out of the vehicle. In some embodiments, the first platform is rotatable in a first circumferential direction relative to the plate and a second circumferential direction, opposite the first circumferential direction, and the first platform is rotatable in a third circumferential direction relative to the plate and a fourth circumferential direction, opposite the third circumferential direction. In some embodiments, the second platform is rotatable in a fifth circumferential direction relative to the plate and a sixth circumferential direction, opposite the fifth circumferential direction.

(11) According to aspects illustrated herein, there is provided an extendable lift assembly for a motor vehicle, comprising a sliding rail assembly, including a plate slidably connected to the vehicle, and a knuckle arm pivotably connected to the plate, a first platform pivotably connected to the knuckle arm, and a second platform extending from and pivotably connected to the plate.

(12) According to aspects illustrated herein, there is provided an extendable lift assembly, comprising a sliding rail assembly, including a plate and a knuckle arm pivotably connected to the plate, and a first platform pivotably connected to the knuckle arm.

(13) In some embodiments, the extendable lift assembly further comprises a static rail operatively arranged to be connected to a vehicle, wherein the sliding rail assembly is slidably connected to the static rail. In some embodiments, the extendable lift assembly further comprises a motor including a gear, wherein the motor and the gear are operatively arranged to displace the sliding rail assembly with respect to the static rail. In some embodiments, the extendable lift assembly further comprises a rack connected to the plate, wherein the gear is operatively arranged to engage the rack. In some embodiments, the extendable lift assembly further comprises a lever operatively arranged to engage and disengage the gear from the rack. In some embodiments, the first platform is connected to the knuckle arm via a shaft. In some embodiments, the first platform is further connected to the knuckle arm via a rod, the rod being pivotably connected to the knuckle arm and pivotably and slidably connected to the first platform. In some embodiments, the knuckle arm is connected to the plate via a shaft. In some embodiments, the first platform is rotatable in a first circumferential direction relative to the knuckle arm and a second circumferential direction, opposite the first circumferential direction, and the knuckle arm is rotatable in a third circumferential direction relative to the plate and a fourth circumferential direction, opposite the third circumferential direction. In some embodiments, the extendable lift assembly further comprises a second platform connected to the sliding rail assembly. In some embodiments, the second platform is hingedly connected to the plate. In some embodiments, the extendable lift assembly further comprises a bridge plate hingedly connected to the first platform. In some embodiments, the extendable lift assembly further comprises an actuator connected at a first end to the bridge plate and at a second end to the first platform, the actuator operatively arranged to circumferentially displace both the bridge plate and the platform. In some embodiments, the extendable lift assembly further comprises a locking plate slidably connected to the plate, and a pin connected to the knuckle arm, wherein the locking plate is operatively arranged to engage the pin to non-rotatably connect the knuckle arm and the plate. In some embodiments, the knuckle arm comprises an arm pivotably connected thereto, and the static rail comprises a slot, wherein the arm is operatively arranged to engage the slot to circumferentially displace the knuckle arm.

(14) According to aspects illustrated herein, there is provided an extendable lift assembly for a vehicle operatively arranged to extend from proximate a floor of the vehicle to proximate a ground surface, the extendable lift assembly comprising a static rail connected to the vehicle, a sliding rail assembly, including a plate slidably connected to the static rail, and a knuckle arm pivotably connected to the plate, a first platform pivotably connected to the knuckle arm, and a drive mechanism, including a motor connected to the static rail, and a gear connected to the motor, the gear operatively arranged to displace the sliding rail assembly with respect to the static rail.

(15) In some embodiments, the extendable lift assembly further comprises a rack connected to the plate, wherein the gear is operatively arranged to engage the rack. In some embodiments, the first platform is rotatable in a first circumferential direction relative to the plate and a second circumferential direction, opposite the first circumferential direction, and the first platform is rotatable in a third circumferential direction relative to the plate and a fourth circumferential direction, opposite the third circumferential direction. In some embodiments, the extendable lift assembly further comprises a bridge plate hingedly connected to the first platform, the bridge plate rotatable in a first circumferential direction relative to the plate and a second circumferential direction, opposite the first circumferential direction. In some embodiments, the extendable lift

assembly further comprises a second platform connected to the sliding rail assembly.

(16) According to aspects illustrated herein, there is provided an extendable lift assembly for a vehicle, comprising a sliding rail assembly, including a plate slidably connected to the vehicle, and a knuckle arm pivotably connected to the plate, and at least one first platform pivotably connected to the sliding rail assembly.

(17) These and other objects, features, and advantages of the present disclosure will become readily apparent upon a review of the following detailed description of the disclosure, in view of the drawings and appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Various embodiments are disclosed, by way of example only, with reference to the accompanying schematic drawings in which corresponding reference symbols indicate corresponding parts, in which:
- (2) FIG. 1A is a perspective view of an extendable lift assembly, in an extended position;
- (3) FIG. 1B is a perspective view of the extendable lift assembly shown in FIG. 1A;
- (4) FIG. 2 is a detail view of the extendable lift assembly taken generally at Detail 2 in FIG. 1A;
- (5) FIG. 3 is a front elevational view of the extendable lift assembly shown in FIG. 1A, in a fully stowed position;
- (6) FIG. 4 is a left side elevational view of an extendable lift assembly, in a partially extended position;
- (7) FIG. 5 is a left side elevational view of the extendable lift assembly shown in FIG. 4, in a fully extended position;
- (8) FIG. 6 is a left side elevational view of the extendable lift assembly shown in FIG. 4, in a fully stowed position;
- (9) FIG. 7 is a partial perspective view of the extendable lift assembly shown in FIG. 1A;
- (10) FIG. 8 is a partial perspective view of the extendable lift assembly shown in FIG. 1A; and
- (11) FIG. 9 is a partial side elevational view of an extendable lift assembly with the bridge plate in an extended position.

DETAILED DESCRIPTION

- (12) At the outset, it should be appreciated that like drawing numbers on different drawing views identify identical, or functionally similar, structural elements. It is to be understood that the claims are not limited to the disclosed aspects.
- (13) Furthermore, it is understood that this disclosure is not limited to the particular methodology, materials and modifications described and as such may, of course, vary. It is also understood that the terminology used herein is for the purpose of describing particular aspects only, and is not intended to limit the scope of the claims.
- (14) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood to one of ordinary skill in the art to which this disclosure pertains. It should be understood that any methods, devices or materials similar or equivalent to those described herein can be used in the practice or testing of the example embodiments. The assembly of the present disclosure could be driven by hydraulics, electronics, pneumatics, actuators (e.g., screw drive), and/or springs.
- (15) It should be appreciated that the term “substantially” is synonymous with terms such as “nearly,” “very nearly,” “about,” “approximately,” “around,” “bordering on,” “close to,” “essentially,” “in the neighborhood of,” “in the vicinity of,” etc., and such terms may be used interchangeably as appearing in the specification and claims. It should be appreciated that the term “proximate” is synonymous with terms such as “nearby,” “close,” “adjacent,” “neighboring,”

“immediate,” “adjoining,” etc., and such terms may be used interchangeably as appearing in the specification and claims. The term “approximately” is intended to mean values within ten percent of the specified value.

(16) It should be understood that use of “or” in the present application is with respect to a “non-exclusive” arrangement, unless stated otherwise. For example, when saying that “item x is A or B,” it is understood that this can mean one of the following: (1) item x is only one or the other of A and B; (2) item x is both A and B. Alternately stated, the word “or” is not used to define an “exclusive or” arrangement. For example, an “exclusive or” arrangement for the statement “item x is A or B” would require that x can be only one of A and B. Furthermore, as used herein, “and/or” is intended to mean a grammatical conjunction used to indicate that one or more of the elements or conditions recited may be included or occur. For example, a device comprising a first element, a second element and/or a third element, is intended to be construed as any one of the following structural arrangements: a device comprising a first element; a device comprising a second element; a device comprising a third element; a device comprising a first element and a second element; a device comprising a first element and a third element; a device comprising a first element, a second element and a third element; or, a device comprising a second element and a third element.

(17) Moreover, as used herein, the phrases “comprises at least one of” and “comprising at least one of” in combination with a system or element is intended to mean that the system or element includes one or more of the elements listed after the phrase. For example, a device comprising at least one of: a first element; a second element; and a third element, is intended to be construed as any one of the following structural arrangements: a device comprising a first element; a device comprising a second element; a device comprising a third element; a device comprising a first element and a second element; a device comprising a first element and a third element; a device comprising a first element, a second element and a third element; or a device comprising a second element and a third element. A similar interpretation is intended when the phrase “used in at least one of:” is used herein. Furthermore, as used herein, “and/or” is intended to mean a grammatical conjunction used to indicate that one or more of the elements or conditions recited may be included or occur. For example, a device comprising a first element, a second element and/or a third element, is intended to be construed as any one of the following structural arrangements: a device comprising a first element; a device comprising a second element; a device comprising a third element; a device comprising a first element and a second element; a device comprising a first element and a third element; a device comprising a first element, a second element and a third element; or a device comprising a second element and a third element.

(18) By “non-rotatably connected” elements, we mean that: the elements are connected so that whenever one of the elements rotate, all the elements rotate; and relative rotation between the elements is not possible. Radial and/or axial movement of non-rotatably connected elements with respect to each other is possible, but not required.

(19) Referring now to the figures, FIG. 1A is a perspective view of extendable lift assembly **10**, in a partially extended position. FIG. 1B is a perspective view of extendable lift assembly **10**. FIG. 2 is a detail view of extendable lift assembly **10** taken generally at Detail 2. FIG. 3 is a front elevational view of extendable lift assembly **10**, in a fully stowed position. Extendable lift assembly **10** generally comprises static rail **30**, sliding rail assembly **40**, and at least one platform (e.g., platform **80** and/or platform **110**). In some embodiments, extendable lift assembly **10** further comprises motor assembly **20**. The following description should be read in view of FIGS. 1-9.

(20) Static rail **30** is operatively arranged to be connected to the vehicle, for example, the B pillar of the vehicle. Static rail **30** comprises channel **32** and channel **34** (see FIG. 3). Static rail **30** is operatively arranged to slidably engage sliding rail assembly **40**. In some embodiments, static rail **30** further comprises inserts **48** and **50**. Inserts **48** and **50** are arranged in channels **32** and **34**, respectively. Inserts **48** and **50** comprise a material that allows better sliding engagement between sliding rail assembly **40** and static rail **30**. For example, inserts **48** and **50** may be fixedly secured in

channels **32** and **34**, respectively, and may comprise a polymer with a low coefficient of friction. Inserts **48** and **50** may be connected to static rail **30** via any suitable means, for example, rivets, screws, bolts, welding, soldering, adhesives, press fit, etc.

(21) Sliding rail assembly **40** comprises plate **42**, plate **44**, rack **46**, and knuckle arm **60**. Plate **42** is slidably engaged with static rail **30**, and specifically inserts **48** and **50**. Plate **44** is connected to plate **42**. In some embodiments, plate **42** and plate **44** are integrally formed. Rack **46** is connected to plate **44**. Rack **46** comprises a plurality of teeth operatively arranged to engage gear **26**, as will be described in greater detail below. Knuckle arm **60** is pivotably connected to plate **42** via blocks **66** and **68** and shaft **64** (see FIG. 8). Knuckle arm **60** and its various components will be described in greater detail below. It should be appreciated that in some embodiments, sliding rail assembly **40** comprises a plurality of telescoping members that extend and retract with respect to static rail **30**. Sliding rail assembly **40** is operatively arranged to extend in axial direction AD1 to extend platform **80** and/or platform **110** to the user. For example, sliding rail assembly **40** may extend in axial direction AD1 until platform **80** is flush with ground surface **1**. Once a user has boarded platform **80** or platform **110**, sliding rail assembly **40** is operatively arranged to retract in axial direction AD2 to carry the user into the vehicle. The main purpose of the extendable lift assembly **10** is to allow users to bypass stair(s) **6** of the vehicle. As such, sliding rail assembly **40** extends and retracts platform **80** and/or platform **110** over stair(s) **6**.

(22) In some embodiments, extendable lift assembly **10** further comprises motor assembly **20**. Motor assembly **20** is connected to static rail **30** and is operatively arranged to drive sliding rail assembly **40**. Motor assembly **20** comprises motor housing **22**, motor **24**, and gear **26**. The output shaft of motor **24** is non-rotatably connected to gear **26**. Gear **26** is operatively arranged to engage rack **46** of sliding rail assembly. As motor **24** rotates in a first circumferential direction, gear **26** drives rack **46** in a first linear direction thereby extending sliding rail assembly **40** and platform **80** (and/or platform **110**) out of the vehicle and toward ground surface **1** (i.e., in axial direction AD1). As motor **24** rotates in a second circumferential direction, opposite the first circumferential direction, gear **26** drives rack **46** in a second linear direction, opposite the first linear direction, thereby retracting sliding rail assembly **40** and platform **80** (and/or platform **110**) back in the vehicle (i.e., in axial direction AD2). In some embodiments, motor assembly **20** further comprises gear engagement lever **28**. Gear engagement lever **28** is connected to gear **26** and is operatively arranged to engage gear **26** with rack **46** and disengage gear **26** from rack **46**. For example, in a disengaged mode (not shown), gear engagement lever **28** separates gear **26** from rack **46** such that the teeth of gear **26** are fully disengaged from the teeth of rack **46**. In an engaged mode (as best shown in FIG. 2), gear engagement lever **28** displaces gear **26** toward rack **46** such that the teeth of gear **26** are engaged with the teeth of rack **46**. As such, if the power of the vehicle fails and motor **24** cannot be activated, gear **26** can be disengaged from rack **46** and sliding rail assembly **40** can be manually retracted within the vehicle. In some embodiments, and as shown in FIGS. 1A-2, motor **24** is mounted underneath static rail **30**. Specifically, a plate of motor assembly **20** is fixedly secured to static rail **30** and motor **24** is fixedly secured to the plate. The shaft of motor **24** extends through the plate and connected to gear **26**. Gear **26** engages rack **46** through an aperture in static rail **30** (see FIG. 2).

(23) In some embodiments, motor **24** is controlled by a circuit (e.g., a high amperage circuit board), is used to displace sliding rail assembly **40** in axial directions AD1 and AD2. The circuit, for example, can be arranged as a main controller for lift assembly **10**. In some embodiments, an encoder is arranged on gear **26** to communicate with the circuit to provide position data regarding sliding rail assembly **40** (i.e., is sliding rail assembly in the fully retracted position, the partially extended position, or the fully extended position). In some embodiments, communication between motor **24** and its control circuit, and the main controller occurs via Controller Area Network (CAN bus). In some embodiments, power and signal wires are run down sliding rail assembly **40** using drag chain or conduit **47**.

(24) It should be appreciated that although the figures show the use of an electric motor as the drive mechanism, one having ordinary skill in the art would appreciate that any drive mechanism suitable for extending and retracting sliding rail assembly **40** and platform **80** (and/or platform **110**) out of and back into a vehicle can be used (e.g., a hydraulic drive mechanism, a pneumatic drive mechanism, a manual drive mechanism, an actuator (e.g., screw drive), etc.).

(25) Platform **80** is pivotably connected to knuckle arm **60** and is operatively arranged to rotate from a use position, as shown in FIGS. **1A-B**, to a stowed position, as shown in FIG. **3**. As shown in FIGS. **1A-B**, in the use position, platform **80** is substantially parallel to vehicle floor **4** and ground surface **1**. Platform **80** comprises plate **82** and is pivotably connected to knuckle arm **60** via shaft **84** and rod **86**. Specifically, shaft **84** extends through platform **80** and knuckle arm **60**. Knuckle arm **60** may comprise bushings and/or bearings in section **72** for rotatable connection to shaft **84**. Similarly, platform **80** may comprise bushings and/or bearings for rotatable connection to shaft **84**. In some embodiments, shaft **84** is rotatably connected to platform **80** and non-rotatably connected to knuckle arm **60**. In some embodiments, shaft **84** is non-rotatably connected to platform **80** and rotatably connected to knuckle arm **60**. In some embodiments, shaft **84** is rotatably connected to platform **80** and rotatably connected to knuckle arm **60**. Rod **86** extends from knuckle arm **60** to plate **80**, and is connected to plate **82**. Rod **86** is rotatably connected to knuckle arm **60** via shaft **90** at a first end and rotatably and slidably connected to plate **82** at a second end, namely, channel **83**. As shown in FIG. **1**, for example, in the full use position, the second end of rod **86** abuts against a first end of channel **83**. In the fully stowed position as shown in FIG. **3**, for example, the second end of rod **86** abuts against the second end of channel **83**. Rod **86** provides support to platform **80**, similar to that of a truss. Rod **86** further imposes a rotational limit on platform **80**, such that platform **80** may only rotate to a position that is substantially parallel to vehicle floor **4** and ground surface **1**. It should be appreciated that the length of rod **86** is adjustable (i.e., like a turnbuckle connecting rod).

(26) Platform **80** is arranged to be rotated to the stowed and use positions, automatically, via actuator **85**, as will be described in greater detail below. However, it should be appreciated that platform **80** may be manually lowered by a user to the use position or lifted to the stowed position. Additionally, platform **80** may be displaced between the use and stowed position via any suitable means, for example, a motor, hydraulics, pneumatics, etc. Platform **80** may be used to lift a standing user into and out of a vehicle. Platform **80** can be used to lift a wheelchair into and out of a vehicle. To shift from the fully stowed position (FIG. **3**) to the use position (FIGS. **1A-B**), knuckle arm **60** rotates in circumferential direction **CD3** and platform **80** rotates in circumferential direction **CD1**. To shift from the use position (FIGS. **1A-B**) to the fully stowed position (FIG. **3**), platform **80** rotates in circumferential direction **CD2** and knuckle arm **60** rotates in circumferential direction **CD4**.

(27) In some embodiments, extendable lift assembly **10** further comprises bridge plate **100**. Bridge plate **100** is hingedly connected to platform **80** via hinge **102**. Bridge plate **100** is operatively arranged to bridge the gap (if any) between platform **80** and vehicle floor **4**, as shown in FIG. **9**. Bridge plate **100** is operatively arranged to rotate in circumferential direction **CD2** from the stowed position shown in FIG. **7** to the use position shown in FIG. **9**. Once sliding rail assembly retracts platform **80** up to vehicle floor **4**, actuator **85** can rotate bridge plate **100** in circumferential direction **CD2** to the use position, shown in FIG. **9**, and the user can step from or roll off of platform **80** onto vehicle floor **4**. In some embodiments, bridge plate **100** is manually rotated to the use and stow position. In some embodiments, bridge **100** further comprises bumper **76** to prevent platform **80** from engaging bridge **100** (i.e., limit the rotational position of bridge plate **100**). For example, as actuator **85** rotates platform **80** in circumferential **CD2**, platform **80** abuts against bumper **76** preventing the two (metal) plates from touching and being damaged. In some embodiments, bridge plate **100** is telescopingly engaged with platform **80**. In such embodiments, bridge plate **100** extends out of an inner cavity of platform **80** to a use position, and retracts into the

inner cavity of platform **80** in a stowed position.

(28) Actuator **85** is pivotably connected at a first end to platform **80** and pivotably connected at a second end to bridge plate **100**. In some embodiments, actuator **85** is an electric linear actuator. It should be appreciated, however, that actuator **85** may comprise any actuator suitable for displacing both platform **85** and bridge plate **100** to the use and stowed position, for example, hydraulic, pneumatic, mechanical, etc. In some embodiments, platform **85** is controlled using microcontroller **200** arranged within platform **80** (see FIG. 7). Microcontroller **200** communicates with the main controller in order to operate actuator **85** (i.e., microcontroller **200** runs actuator **85** to extend and retract). In some embodiments, actuator **85** comprises an encoder which provides position data to microcontroller **200** and main controller **200**. As previously described, rod **86** is slidably and pivotably connected to platform **80** and limits the rotation of platform **80** in circumferential direction CD1. Specifically, rod **86** prevents platform **80** from displacing more than substantially parallel to ground surface **1** (see FIGS. 1A-B). Once platform is in its fully extended or use position (FIGS. 1A-B), actuator **85** can be more fully extended which will force bridge plate **100** to displace in circumferential direction CD2 toward vehicle floor **4**. Actuator **85** will shut off once bridge plate **100** is fully extended and flush with vehicle floor **4** (see FIG. 9). Once the user steps into the vehicle from platform **80** and bridge plate **100**, microcontroller **200** begins retracting bridge plate **100** in circumferential direction CD1 by simply retracting actuator **85**. Once bridge plate **100** displaces to the fully stowed position it engages stopper **70** (see FIGS. 1B and 7). Stopper **70** is non-rotatably connected to section **72** of knuckle arm and limits displacement of bridge plate **100** in circumferential direction CD1. Once bridge plate **100** is engaged with stopper **70**, further retraction of actuator **85** causes platform **80** to rotate in circumferential direction CD2. Actuator **85** will displace platform **80** in circumferential direction CD2 until platform **80** engages stopper **76**. (29) Platform **110** is pivotably connected to sliding rail assembly **40**. In some embodiments, platform **110** is pivotably connected to plate **42** via hinge **112**. Hinge **112** may comprise a folding shelf bracket or an equivalent thereof, wherein platform **110** is capable of being raised to the use position (as shown in FIGS. 1A-B) and locked, as well as collapsed to a stowed position (not shown). In some embodiments, hinge **112** comprises one or more folding shelf brackets that lock at 90°, such that platform **110** is substantially parallel to vehicle floor **4** and ground surface **1**. Platform **110** can be used to lift a seated user into and out of a vehicle.

(30) FIG. 4 is a left side elevational view of extendable lift assembly **10**. As shown in FIG. 4, in some embodiments extendable lift assembly **10** may comprise handle assembly **150** operatively arranged to provide a support rail for a user standing on platform **80**. Handle assembly **150** comprises post **152** connected to sliding rail assembly **40** and bar **154** connected to post **152**. In some embodiments, bar **154** is connected to post **152** via bracket **156**. Bracket **156** allows bar **154** to be collapsed in circumferential direction CD1 when not in use, and rotated in circumferential direction CD2 and locked in place (as shown in FIG. 4) for use. It should be appreciated that bar **154** can be positioned for use or stowed either manually by a user or automatically (e.g., via an electric motor, hydraulics, pneumatics, actuators (e.g., screw drives, etc.).

(31) FIG. 5 is a left side elevational view of extendable lift assembly **10**. FIG. 6 is a left side elevational view of extendable lift assembly **10**, in a stowed position. As shown in FIGS. 5 and 6, in some embodiments extendable lift assembly **10** may comprise handle assembly **160** operatively arranged to provide a support rail for a user standing on platform **80**. Handle assembly **160** comprises post **162** connected to sliding rail assembly **40** and bar **164** connected to post **152**. In some embodiments, bar **164** is telescopingly connected with post **162**. In the use position, as shown in FIG. 5, bar **164** is extended from post **162**. In the stowed position, as shown in FIG. 6, bar **164** is retracted substantially within post **162**. It should be appreciated that bar **164** can be positioned for use or stowed either manually by a user or automatically (e.g., via an electric motor, hydraulics, pneumatics, actuators (e.g., screw drive), etc.). Also shown in FIGS. 5 and 6 is the use position and the stowed position of platform **80**. To shift from the use position shown in FIG. 5 to the stowed

position shown in FIG. 6, platform **80** is rotated in circumferential direction CD2 with respect to shaft **84**. Then platform **80**, and knuckle arm **60**, is rotated in circumferential direction CD4 with respect to section **62** and shaft **64** (see FIG. 8) therein. To shift from the stowed position shown in FIG. 6 to the use position shown in FIG. 5, platform **80**, and knuckle arm **60**, is rotated in circumferential direction CD3 with respect to section **62** and shaft **64** (see FIG. 8) therein. Then platform **80** is rotated in circumferential direction CD1.

(32) FIG. 7 is a partial perspective view of extendable lift assembly **10**. FIG. 8 is a perspective view of knuckle arm **60**. FIG. 9 is a side elevational view of extendable lift assembly **10**. The following description should be read in view of FIGS. 1A-9.

(33) Knuckle arm **60** is pivotably connected to sliding rail assembly **40** and platform **80**. Knuckle arm **60** comprises section **62** and section **72**. Knuckle arm **60** further comprises block **66** and block **68**, which are connected to sliding rail assembly **40**. In some embodiments, blocks **66** and **68** are fixedly secured to plate **42** via any suitable means, for example, bolts, rivets, screws, adhesives, welding, soldering, interference fit, etc. Shaft **64** extends through section **62** and is connected to blocks **66** and **68**. In some embodiments, shaft **64** is non-rotatably connected to blocks **66** and **68** and rotatably connected to section **62**. In such embodiments, knuckle arm **60** may further comprise a pull pin operatively arranged to lock knuckle arm **60** with respect to shaft **64**, for example, in the stowed or the use position. In some embodiments, section **62** further comprises one or more bearings to encourage rotational motion between section **62** and shaft **64**. In some embodiments, shaft **64** is rotatably connected to blocks **66** and **68** and non-rotatably connected to section **62**.

(34) Section **72** is rotatably connected to platform **80** via shaft **84**. In some embodiments, shaft **84** is rotatably connected to section **72** and non-rotatably connected to platform **80**. In such embodiments, knuckle arm **60** may further comprise a pull pin operatively arranged to lock knuckle arm **60** with respect to shaft **84**, for example, in the stowed or the use position. In some embodiments, section **72** further comprises one or more bearings to encourage rotational motion between section **72** and shaft **84**. In some embodiments, shaft **84** is rotatably connected to platform **80** and non-rotatably connected to section **72**. As previously described, stopper **70** is non-rotatably connected to knuckle arm **60**, specifically section **72**, to limit displacement of bridge plate **100** in circumferential direction CD1. Once bridge plate **100** is engaged with stopper **70**, further retraction of actuator **85** causes platform **80** to displace in circumferential direction CD2.

(35) As best shown in FIG. 2, lift assembly **10** further comprises pin **174** and locking plate **170**. Pin **174** is fixedly secured to knuckle arm **60**. Locking plate **170** is slidably engaged with plate **44** and comprises slot **172**, spring **176**, and surface **178**. In the locked position, and as shown in FIG. 2, locking plate **170** is displaced in axial direction AD3 and slot **172** is fully engaged with pin **174** thereby rotatably locking knuckle arm **60** with respect to plate **44** (i.e., sliding rail assembly **40**). In this fully locked position, knuckle arm **60** cannot be displaced in circumferential direction CD3 or CD4. In an unlocked position (not shown), locking plate **170** is displaced in axial direction AD4 such that slot **172** completely disengages pin **174** thereby allowing knuckle arm **60** to be displaced in circumferential direction CD3 and/or CD4. In some embodiments, as sliding rail assembly **40** is displaced in axial direction AD2 to be stowed, surface **178** engages insert **50** (or another component of sliding rail assembly **40**), thereby forcing locking plate **170** in axial direction AD4 and out of engagement with pin **174**, allowing rotational displacement of knuckle arm **60** with respect to sliding rail assembly **40**. Spring **176** connects locking plate **170** and plate **44** and biases locking plate **170** in axial direction AD3, or toward the locked position. It should be appreciated that when a passenger is standing or sitting on platform **80** and/or platform **110**, locking plate **170** will be in the locked position (i.e., fully engaged with pin **174**) due to the weight of the passenger forcing locking plate **170** in axial direction AD3. Locking plate **170** is operatively arranged to automatically lock and unlock knuckle arm **60** rotationally with respect to sliding rail assembly **40**.

(36) Knuckle arm **60** further comprises arm **180** which is pivotably connected to knuckle arm **60**

via shaft **182**. Arm **180** is specifically connected to section **62** and may include a slide plate and/or a block. Arm **180** is operatively arranged to engage slot **190** in static rail **30** to displace knuckle arm **60** in circumferential direction **CD4**. To put lift assembly **10** in the stowed position, sliding rail assembly **40** is displaced in axial direction **AD2**. Knuckle arm **60** is rotationally unlocked from sliding rail assembly **40** as locking plate **170** is displaced in axial direction **AD4**, thereby disengaging slot **172** from pin **174**. Arm **180** then engages slot **190**. As sliding rail assembly **40** continues displacing in axial direction **AD2**, arm **180** remains behind pulling on shaft **182** thereby displacing knuckle arm **60** in circumferential direction **CD4** to the stowed position (i.e., arm **180** acts as a lever arm on section **62**). Arm **180** is further connected to plate **44** via spring **186**. To put lift in the use position, as shown in FIGS. **1A-2**, sliding rail assembly **40** is displaced in axial direction **AD1**. As arm **180** disengages from slot **190**, spring **186** biases arm toward plate **44** thereby displacing knuckle arm **60** in circumferential direction **CD3**. In some embodiments, arm **180** further comprises stopper **184** to limit displacement of arm **180**.

(37) It will be appreciated that various aspects of the disclosure above and other features and functions, or alternatives thereof, may be desirably combined into many other different systems or applications. Various presently unforeseen or unanticipated alternatives, modifications, variations, or improvements therein may be subsequently made by those skilled in the art which are also intended to be encompassed by the following claims.

REFERENCE NUMERALS

(38) **1** Ground surface **4** Vehicle floor **6** Step(s) **20** Lift assembly **22** Motor assembly **22** Motor housing **24** Motor **26** Gear **28** Gear engagement lever **30** Static rail **32** Channel **34** Channel **40** Sliding rail assembly **42** Plate **44** Plate **46** Rack **48** Insert **50** Insert **60** Knuckle arm **62** Section **64** Shaft **66** Block **68** Block **70** Bumper **72** Section **76** Bumper **80** Platform **82** Plate **83** Channel **84** Shaft **85** Actuator **86** Rod **88** Shaft **90** Shaft **100** Bridge plate **102** Hinge **110** Platform **112** Hinge **150** Post **152** Bar **154** Bracket **160** Handle assembly **162** Post **164** Bar **170** Locking plate **172** Slot **174** Pin **176** Spring **178** Surface **180** Arm or block **182** Shaft **184** Stopper **186** Spring **190** Slot **200** Microcontroller **AD1** Axial direction **AD2** Axial direction **AD3** Axial direction **AD4** Axial direction **CD1** Circumferential direction **CD2** Circumferential direction **CD3** Circumferential direction **CD4** Circumferential direction

Claims

1. An extendable lift assembly for a vehicle, comprising: a sliding rail assembly, including: a plate slidably connected to the vehicle; and an arm pivotably connected to the plate; a first platform pivotably connected to the arm; and a second platform extending from and pivotably connected to the plate, wherein the second platform is a seat for a user and is spaced apart from the first platform along the plate.
2. The extendable lift assembly as recited in claim 1, further comprising a static rail operatively arranged to be fixedly secured to the vehicle, wherein the plate is slidingly connected to the static rail.
3. The extendable lift assembly as recited in claim 2, further comprising a motor including a gear, wherein the motor and the gear are operatively arranged to displace the plate with respect to the static rail.
4. The extendable lift assembly as recited in claim 3, further comprising a rack connected to the plate, wherein the gear is operatively arranged to engage the rack.
5. The extendable lift assembly as recited in claim 4, further comprising a lever operatively arranged to engage and disengage the gear from the rack.
6. The extendable lift assembly as recited in claim 2, wherein: the arm comprises an engaging member pivotably connected thereto; and the static rail comprises a slot, wherein the engaging member is operatively arranged to engage the slot to circumferentially displace the arm.

7. The extendable lift assembly as recited in claim 1, wherein the first platform is connected to the arm via a shaft.
 8. The extendable lift assembly as recited in claim 7, wherein the first platform is further connected to the arm via a rod, the rod being pivotably connected to the arm and pivotably and slidably connected to the first platform.
 9. The extendable lift assembly as recited in claim 1, wherein the arm is connected to the plate via a shaft.
 10. The extendable lift assembly as recited in claim 1, wherein: the first platform is rotatable in a first circumferential direction relative to the arm and a second circumferential direction, opposite the first circumferential direction; and the arm is rotatable in a third circumferential direction relative to the plate and a fourth circumferential direction, opposite the third circumferential direction.
 11. The extendable lift assembly as recited in claim 1, further comprising a bridge plate hingedly connected to the first platform.
 12. The extendable lift assembly as recited in claim 11, further comprising an actuator connected at a first end to the bridge plate and at a second end to the first platform, the actuator operatively arranged to circumferentially displace both the bridge plate and the first platform.
 13. The extendable lift assembly as recited in claim 1, further comprising: a locking plate slidably connected to the plate; and a pin connected to the arm, wherein the locking plate is operatively arranged to engage the pin to non-rotatably connect the arm and the plate.
 14. The extendable lift assembly as recited in claim 1, further comprising a drive mechanism operatively arranged to displace the plate with respect to the static rail to extend and retract the plate.
 15. An extendable lift assembly for a vehicle operatively arranged to extend from proximate a floor of the vehicle to proximate a ground surface, the extendable lift assembly comprising: a sliding rail assembly, including: a plate slidably connected to the vehicle; and an arm pivotably connected to the plate; a first platform pivotably connected to the arm; a second platform pivotably connected to the plate; and a drive mechanism operatively arranged to displace the sliding rail assembly with respect to the vehicle.
 16. The extendable lift assembly as recited in claim 15, wherein at least one of the first platform and the second platform is operatively arranged to carry a user into and out of the vehicle.
 17. The extendable lift assembly as recited in claim 15, wherein: the first platform is rotatable in a first circumferential direction relative to the plate and a second circumferential direction, opposite the first circumferential direction; and the first platform is rotatable in a third circumferential direction relative to the plate and a fourth circumferential direction, opposite the third circumferential direction.
 18. The extendable lift assembly as recited in claim 17, wherein the second platform is rotatable in a fifth circumferential direction relative to the plate and a sixth circumferential direction, opposite the fifth circumferential direction.
 19. An extendable lift assembly for a motor vehicle, comprising: a sliding rail assembly, including: a plate slidably connected to the vehicle; and an arm pivotably connected to the plate; a first platform extending from and pivotably connected to the arm; and a second platform extending from and pivotably connected to the plate, wherein the second platform is spaced apart from the first platform along the plate.
 20. The extendable lift assembly as recited in claim 19, further comprising a bridge plate hingedly connected to the first platform.
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